

Diffusion models for particle physics simulations

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May 15, 2025

1. Particle physics simulations

2. Diffusion Models

3. Equivariant learning

4. Appendices

Particle physics simulations

Collisions

High energy protons $\xrightarrow{\text{collisions}}$ Non-isolatable elementary particles $\xrightarrow{\text{radiate}}$ Parton shower
 $\xrightarrow[\text{hadronization}]{\text{cooling}}$ Jet

Initial particles $\rightarrow$ Jet constituents

Simulation-based hypothesis testing

$$p(\beta|o) = \frac{p(o|\beta)p(\beta)}{\int p(\beta')p(o|\beta') d\beta'}$$

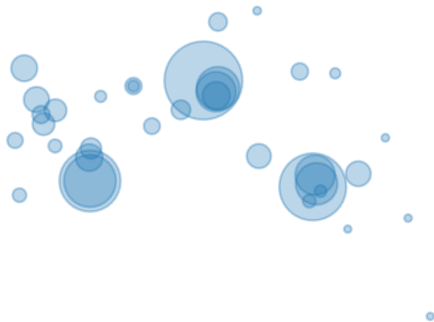
- Input: z_0, β
- Latent states: $z_i \sim p_i(z_i|\beta, z_{<i})$
- Output: $o \sim p(o|\beta, z)$

$$p(o|\beta) = \int p(o, \beta|\beta) dz$$

Limitations

- Fit simplified phenomenological models to parton shower, hadronization
- Slow: 10 s / jet
- One-shot generation: 3-5 OoM speedup!

Representing jets



Gouskos and Qu 2018

Particle clouds + feature vectors

Particle physics simulations | Diffusion models for simulations

Diffusion Models

Generative modeling

Given: $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N$ from $p_{\text{data}}(\mathbf{x})$

Learn f_θ s.t.

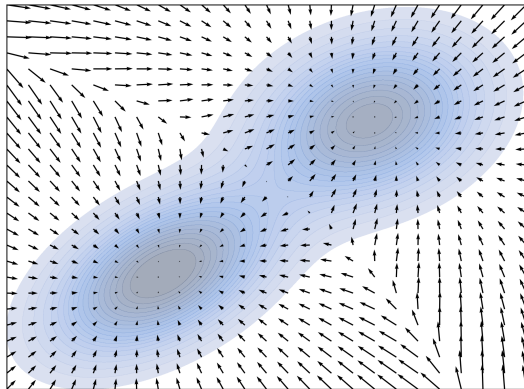
$$p_\theta(\mathbf{x}) = \frac{e^{f_\theta(\mathbf{x})}}{Z_\theta}$$

$$Z_\theta = \int e^{f_\theta(\mathbf{x})} d\mathbf{x}$$

Score functions

$$p_{\theta}(\mathbf{x}) = \frac{e^{f_{\theta}(\mathbf{x})}}{Z_{\theta}}$$

$$\begin{aligned}\nabla_{\mathbf{x}} \log p_{\theta}(\mathbf{x}) &= \nabla_{\mathbf{x}} f_{\theta}(\mathbf{x}) - \nabla_{\mathbf{x}} \log Z_{\theta} \\ &= \nabla_{\mathbf{x}} f_{\theta}(\mathbf{x}) = s_{\theta}(\mathbf{x})\end{aligned}$$

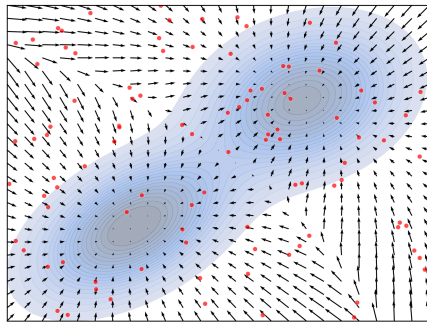


Sampling

$$\mathbf{x}_{t+1} = \mathbf{x}_t + \alpha \cdot s_{\theta}(\mathbf{x}_t)$$

$$x_0 \sim \pi_{\text{init}}$$

$$t = 0, 1, \dots, T \rightarrow \infty$$



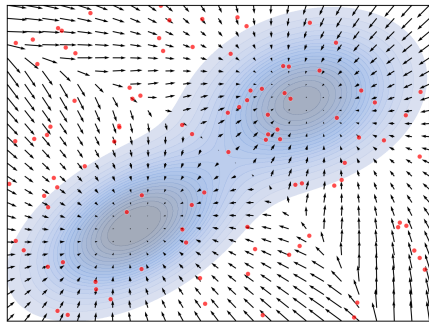
Sampling with Langevin dynamics

$$\mathbf{x}_{t+1} = \mathbf{x}_t + \alpha \cdot s_{\theta}(\mathbf{x}_t) + \alpha \epsilon \cdot \mathbf{z}_t$$

$$x_0 \sim \pi_{\text{init}}$$

$$\mathbf{z}_t \sim \mathcal{N}(0, 1)$$

$$t = 0, 1, \dots, T \rightarrow \infty$$



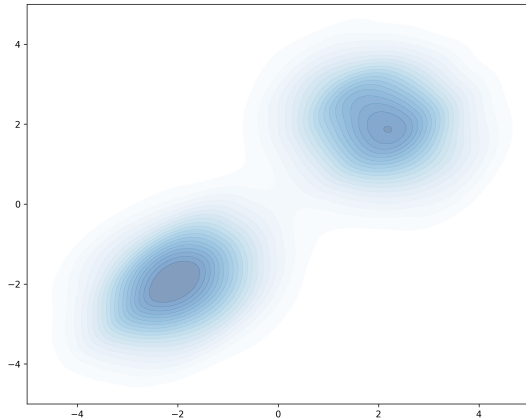
Training objective for score

$$\mathcal{L}_{\text{ESM}}(\theta) = \frac{1}{2} \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \left[\|\nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x}) - s_{\theta}(\mathbf{x})\|_2^2 \right]$$

$$\iff \mathcal{L}_{\text{ISM}}(\theta) = \frac{1}{2} \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \left[s_{\theta}(\mathbf{x})^2 \right] + \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \left[\nabla_{\mathbf{x}} s_{\theta}(\mathbf{x}) \right]$$

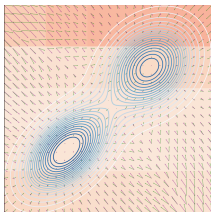
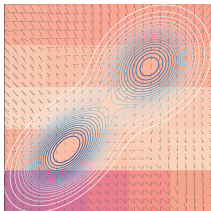
$$\approx \frac{1}{N} \sum_{i=1}^N \left[\frac{1}{2} s_{\theta}(\mathbf{x}_i)^2 + \nabla_{\mathbf{x}_i} s_{\theta}(\mathbf{x}_i) \right]$$

Training



Drawbacks

Field in low-density regions



Scaling

$$\frac{1}{2} \mathbb{E}_{p_{\text{data}}(\mathbf{x})} [s_{\theta}(\mathbf{x})^2] + \mathbb{E}_{p_{\text{data}}(\mathbf{x})} [\nabla_{\mathbf{x}} s_{\theta}(\mathbf{x})]$$

$$\frac{1}{2} \mathbb{E}_{p_{\text{data}}(\mathbf{x})} [s_{\theta}(\mathbf{x})^2] + \mathbb{E}_{p_{\text{data}}(\mathbf{x})} [\text{trace}(\nabla_{\mathbf{x}} s_{\theta}(\mathbf{x}))]$$

Forward propagation Backprops $\propto \text{dim}(\mathbf{x})$

Noise perturbation

$$\tilde{\mathbf{x}} = \mathbf{x} + z$$

$$z \sim \mathcal{N}(0, \sigma^2 I)$$

$$q_\sigma(\tilde{\mathbf{x}}|\mathbf{x}) = \mathcal{N}(\mathbf{x}, \sigma^2 I)$$

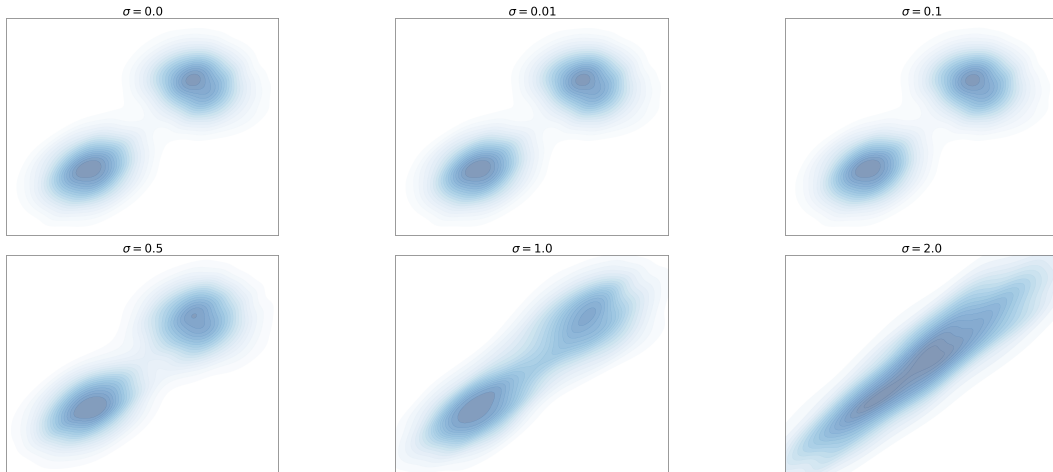
Denoising objective

$$\begin{aligned}\mathcal{L}_{\text{ESM}\sigma}(\theta, \sigma) &= \frac{1}{2} \mathbb{E}_{q_\sigma(\tilde{\mathbf{x}})} \left[\|s_\theta(\tilde{\mathbf{x}}) - \nabla_{\tilde{\mathbf{x}}} \log q_\sigma(\tilde{\mathbf{x}})\|_2^2 \right] \\ \iff \mathcal{L}_{\text{ISM}\sigma}(\theta, \sigma) &= \frac{1}{2} \mathbb{E}_{q_\sigma(\tilde{\mathbf{x}}, \mathbf{x})} \left[\|s_\theta(\tilde{\mathbf{x}}) - \nabla_{\tilde{\mathbf{x}}} \log q_\sigma(\tilde{\mathbf{x}}|\mathbf{x})\|_2^2 \right]\end{aligned}$$

Gradient of the corrupted distribution

$$\begin{aligned}\nabla_{\tilde{\mathbf{x}}} \log q_{\sigma}(\tilde{\mathbf{x}}|\mathbf{x}) &= \nabla_{\tilde{\mathbf{x}}} \log \frac{1}{(2\pi)^{\frac{d}{2}} \sigma^d} e^{-\frac{1}{2\sigma^2} \|\tilde{\mathbf{x}} - \mathbf{x}\|^2} \\&= \nabla_{\tilde{\mathbf{x}}} \log e^{-\frac{1}{2\sigma^2} \|\tilde{\mathbf{x}} - \mathbf{x}\|^2} - \nabla_{\tilde{\mathbf{x}}} \log (2\pi)^{\frac{d}{2}} \sigma^d \\&= -\nabla_{\tilde{\mathbf{x}}} \frac{1}{2\sigma^2} \|\tilde{\mathbf{x}} - \mathbf{x}\|^2 \\&= \frac{\|\tilde{\mathbf{x}} - \mathbf{x}\|}{\sigma^2} \\\therefore \mathcal{L}_{\text{ISM}\sigma}(\theta, \sigma) &= \frac{1}{2} \mathbb{E}_{q_{\sigma}(\tilde{\mathbf{x}}, \mathbf{x})} \left[\left\| s_{\theta}(\tilde{\mathbf{x}}) - \frac{\|\tilde{\mathbf{x}} - \mathbf{x}\|}{\sigma^2} \right\|_2^2 \right]\end{aligned}$$

Multiple noise levels



Multiple noise levels

$$\mathcal{L}(\theta, \{\sigma_i\}_{i=1}^L) = \frac{1}{L} \sum_{i=1}^L \lambda(\sigma_i) \mathbb{E}_{q_{\sigma_i}(\tilde{\mathbf{x}}, \mathbf{x})} \left[\left\| s_{\theta}(\tilde{\mathbf{x}}, \sigma_i) - \frac{\|\tilde{\mathbf{x}} - \mathbf{x}\|}{\sigma^2} \right\|_2^2 \right]$$

Sampling with annealed Langevin dynamics

$$\tilde{\mathbf{x}}_0 \sim \pi_{\text{init}}$$

From $i = 1, 2, \dots, L$ with $\frac{\sigma_1}{\sigma_2} = \dots = \frac{\sigma_{L-1}}{\sigma_L} > 1$

$$\alpha_i = \gamma \frac{\sigma_i^2}{\sigma_L^2}$$

$$\tilde{\mathbf{x}}_t = \tilde{\mathbf{x}}_{t-1} + \alpha_i s_\theta(\tilde{\mathbf{x}}_{t-1}, \sigma_i) + \epsilon \alpha_i \mathbf{z}_t$$

$$\mathbf{z}_t \sim \mathcal{N}(0, 1)$$

Conditional generation

$$\nabla_{\mathbf{x}} \log p(\mathbf{x}|\mathbf{y}) = \frac{p(\mathbf{y}|\mathbf{x})p(\mathbf{x})}{p(\mathbf{y})}$$

$$\begin{aligned}\nabla_{\mathbf{x}} \log p(\mathbf{x}|\mathbf{y}) &= \nabla_{\mathbf{x}} \log p(\mathbf{y}|\mathbf{x}) + \nabla_{\mathbf{x}} \log p(\mathbf{x}) - \nabla_{\mathbf{x}} \log p(\mathbf{y}) \\ &= \nabla_{\mathbf{x}} \log p(\mathbf{x}) + \nabla_{\mathbf{x}} \log p(\mathbf{y}|\mathbf{x}) \\ &= s_{\theta}(\mathbf{x}) + \gamma \nabla_{\mathbf{x}} \log p(\mathbf{y}|\mathbf{x})\end{aligned}$$

Single model for guidance

$$\begin{aligned} p(\mathbf{y}|\mathbf{x}) &= \nabla_{\mathbf{x}} \log p(\mathbf{x}|\mathbf{y}) - \nabla_{\mathbf{x}} \log p(\mathbf{x}) \\ \implies \nabla_{\mathbf{x}} \log p(\mathbf{x}|\mathbf{y}) &= \nabla_{\mathbf{x}} \log p(\mathbf{x}) + \gamma \nabla_{\mathbf{x}} \log p(\mathbf{x}|\mathbf{y}) - \gamma \nabla_{\mathbf{x}} \log p(\mathbf{x}) \\ &= \gamma \nabla_{\mathbf{x}} \log p(\mathbf{x}|\mathbf{y}) + (\gamma - 1) \nabla_{\mathbf{x}} \log p(\mathbf{x}) \\ &\approx \gamma s_{\theta}(\mathbf{x}, \mathbf{y}) - (\gamma - 1) s_{\theta}(\mathbf{x}, \phi) \end{aligned}$$

Final score model

$$s_{theta}(\mathbf{x}, \mathbf{y}, \beta)$$

Equivariant learning

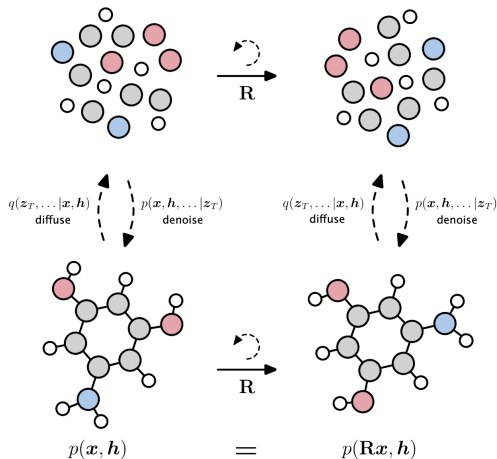
- Equivariant: $f(g(I)) = g'(f(I))$, $g \in G, g' \in G'$
 - Same-equivariant: $f(g(I)) = g(f(I))$
 - Invariant: $f(g(I)) = f(I)$
-
- Images: translation, rotation, reflection
 - Graphs, sets: permutation

Inductive biases

In ML

1. Data augmentation
2. Model architecture

Equivariant diffusion models



Hoogeboom et al. 2022

Lorentz equivariance in diffusion?

$$\text{eg. } \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

(relative velocity $\beta = \frac{v}{c}$, boost factor $\gamma = \frac{1}{(1-\beta^2)^2}$)

Appendices

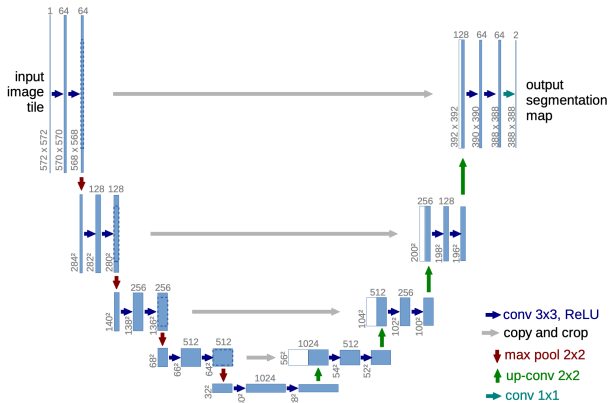
Derivation of divergence objective for score

$$\begin{aligned}& \frac{1}{2} \mathbb{E}_{p_{\text{data}}(\mathbf{x})} [\|\nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x}) - s_{\theta}(\mathbf{x})\|_2^2] \\&= \frac{1}{2} \int p_{\text{data}}(\mathbf{x}) (\nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x}) - \nabla_{\mathbf{x}} f_{\theta}(\mathbf{x}))^2 d\mathbf{x} \\&= \frac{1}{2} \int p_{\text{data}}(\mathbf{x}) (\nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x}))^2 d\mathbf{x} + \frac{1}{2} \int p_{\text{data}}(\mathbf{x}) (\nabla_{\mathbf{x}} f_{\theta}(\mathbf{x}))^2 d\mathbf{x} - \int p_{\text{data}}(\mathbf{x}) \nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x}) \cdot \nabla_{\mathbf{x}} f_{\theta}(\mathbf{x}) d\mathbf{x} \\&\quad \text{Constant in training} \qquad \qquad \qquad \text{Use } \nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x}) = \frac{\nabla_{\mathbf{x}} p_{\text{data}}(\mathbf{x})}{p_{\text{data}}(\mathbf{x})} \\&= C + \frac{1}{2} \int p_{\text{data}}(\mathbf{x}) (\nabla_{\mathbf{x}} f_{\theta}(\mathbf{x}))^2 d\mathbf{x} - \int \nabla_{\mathbf{x}} p_{\text{data}}(\mathbf{x}) \cdot \nabla_{\mathbf{x}} f_{\theta}(\mathbf{x}) d\mathbf{x} \\&\quad \text{Integrate by parts} \\&= C + \frac{1}{2} \int p_{\text{data}}(\mathbf{x}) (\nabla_{\mathbf{x}} f_{\theta}(\mathbf{x}))^2 d\mathbf{x} - p_{\text{data}}(\mathbf{x}) \nabla_{\mathbf{x}} f_{\theta}(\mathbf{x}) \Big|_{-\infty}^{\infty} + \int p_{\text{data}}(\mathbf{x}) \nabla_{\mathbf{x}} \nabla_{\mathbf{x}} f_{\theta}(\mathbf{x}) d\mathbf{x} \\&\quad \text{Boundary values} \rightarrow 0 \\&= C + \frac{1}{2} \mathbb{E}_{p_{\text{data}}(\mathbf{x})} [\nabla_{\mathbf{x}} f_{\theta}(\mathbf{x})^2] + \mathbb{E}_{p_{\text{data}}(\mathbf{x})} [\nabla_{\mathbf{x}}^2 f_{\theta}(\mathbf{x})] = C + \frac{1}{2} \mathbb{E}_{p_{\text{data}}(\mathbf{x})} [s_{\theta}(\mathbf{x})^2] + \mathbb{E}_{p_{\text{data}}(\mathbf{x})} [\nabla_{\mathbf{x}} s_{\theta}(\mathbf{x})]\end{aligned}$$

Derivation of noise-perturbed divergence objective

$$\begin{aligned}
 & \frac{1}{2} \mathbb{E}_{q_\sigma(\tilde{\mathbf{x}})} [\|s_\theta(\tilde{\mathbf{x}}) - \nabla_{\tilde{\mathbf{x}}} \log q_\sigma(\tilde{\mathbf{x}})\|_2^2] \\
 &= \frac{1}{2} \mathbb{E}_{q_\sigma(\tilde{\mathbf{x}})} [s_\theta(\tilde{\mathbf{x}})^2] - \int q_\sigma(\tilde{\mathbf{x}}) s_\theta(\tilde{\mathbf{x}}) \nabla_{\tilde{\mathbf{x}}} \log q_\sigma(\tilde{\mathbf{x}}) d\tilde{\mathbf{x}} + \frac{1}{2} \int q_\sigma(\tilde{\mathbf{x}}) (\nabla_{\tilde{\mathbf{x}}} \log q_\sigma(\tilde{\mathbf{x}}))^2 d\tilde{\mathbf{x}} \\
 & \quad \text{Use } \nabla_{\tilde{\mathbf{x}}} \log q_\sigma(\tilde{\mathbf{x}}) = \frac{\nabla_{\tilde{\mathbf{x}}} q_\sigma(\tilde{\mathbf{x}})}{q_\sigma(\tilde{\mathbf{x}})} \quad \text{Constant in training} \\
 & \quad \int s_\theta(\tilde{\mathbf{x}}) \nabla_{\tilde{\mathbf{x}}} q_\sigma(\tilde{\mathbf{x}}) d\tilde{\mathbf{x}} \\
 & \quad \text{Use } q_\sigma(\tilde{\mathbf{x}}) = \int p_{\text{data}}(\mathbf{x}) p_\sigma(\tilde{\mathbf{x}}|\mathbf{x}) d\mathbf{x} \\
 &= \int s_\theta(\tilde{\mathbf{x}}) \nabla_{\tilde{\mathbf{x}}} \left(\int p_{\text{data}}(\mathbf{x}) q_\sigma(\tilde{\mathbf{x}}|\mathbf{x}) d\mathbf{x} \right) d\tilde{\mathbf{x}} \\
 & \quad \text{Differentiation under integral} \\
 &= \int s_\theta(\tilde{\mathbf{x}}) \left(\int p_{\text{data}}(\mathbf{x}) \nabla_{\tilde{\mathbf{x}}} q_\sigma(\tilde{\mathbf{x}}|\mathbf{x}) d\mathbf{x} \right) d\tilde{\mathbf{x}} = \int \int s_\theta(\tilde{\mathbf{x}}) p_{\text{data}}(\mathbf{x}) q_\sigma(\tilde{\mathbf{x}}|\mathbf{x}) \nabla_{\tilde{\mathbf{x}}} \log q_\sigma(\tilde{\mathbf{x}}|\mathbf{x}) d\mathbf{x} d\tilde{\mathbf{x}} \\
 & \quad \text{Use } \nabla_{\tilde{\mathbf{x}}} q_\sigma(\tilde{\mathbf{x}}|\mathbf{x}) = q_\sigma(\tilde{\mathbf{x}}|\mathbf{x}) \nabla_{\tilde{\mathbf{x}}} \log q_\sigma(\tilde{\mathbf{x}}|\mathbf{x}) \\
 &= \mathbb{E}_{q_\sigma(\tilde{\mathbf{x}}, \mathbf{x})} [s_\theta(\tilde{\mathbf{x}}) \nabla_{\tilde{\mathbf{x}}} \log q_\sigma(\tilde{\mathbf{x}}|\mathbf{x})]
 \end{aligned}$$

U-Net



Source: Ronneberger et al. (2015)